Euclid's Elements — Book I
Proposition 30

CGJteamLab

Straight lines parallel to the same straight line are also parallel to one another.

1 Introduction

Proposition I.30 establishes a structural property of Euclidean parallelism:

$$AB \parallel EF, \qquad CD \parallel EF$$

imply

$$AB \parallel CD.$$

In Euclid's proof the conclusion is obtained by drawing a transversal through the three lines, applying Proposition I.29 twice, using transitivity of angle congruence, and then applying Proposition I.27.

In the Hilbert reconstruction the same theorem is more naturally organized through the uniqueness of a parallel line through a point. This is exactly the content of Hilbert's Group IV.

A formal subtlety must be made explicit. In the project, `Geo.Parallel` is a strict relation: parallel lines are distinct nondegenerate lines with disjoint point-line carriers. Therefore the statement must exclude the case in which the first and third lines are actually the same line.

The formal hypothesis is

$$\mathrm{PointLine}(A, B) \neq \mathrm{PointLine}(C, D).$$

This distinctness is implicit in Euclid's classical diagram, where the three straight lines are drawn as three separate lines.

2 Classical Configuration

Let AB, CD, and EF be three distinct straight lines such that

$$AB \parallel EF$$

and

$$CD \parallel EF.$$

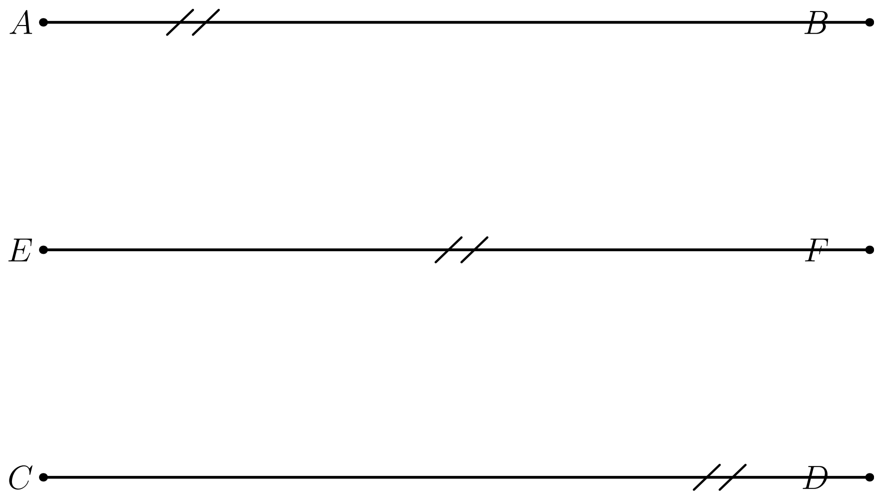

Figure 1: The configuration of Proposition I.30. The two outer lines are each parallel to the same middle line.

The conclusion is

$$AB \parallel CD.$$

3 Euclid's Proof Architecture

Euclid introduces a transversal meeting the three parallel lines.

The first application of Proposition I.29 gives an equality of alternate angles for the first pair of parallel lines. A second application of Proposition I.29 gives the corresponding equality for the second pair.

Thus one obtains a chain of angle congruences

$$\angle AGK \cong \angle GHF \cong \angle GKD.$$

By transitivity,

$$\angle AGK \cong \angle GKD.$$

These are alternate interior angles for the first and third lines. Proposition I.27 therefore gives

$$AB \parallel CD.$$

The classical dependency graph is therefore

$$\begin{array}{cccc}
AB \parallel EF & & CD \parallel EF & \\
\downarrow \text{I.29} & & \downarrow \text{I.29} & \\
\angle AGK \cong \angle GHF & & \angle GHF \cong \angle GKD & \\
& \searrow & \downarrow & \swarrow \\
& & \angle AGK \cong \angle GKD & \\
& & \downarrow \text{I.27} & \\
& & AB \parallel CD. &
\end{array}$$

4 Hilbert Reconstruction

The Hilbert reconstruction isolates the more general structural theorem

`hilbert_parallel_transitive_distinct`

with the mathematical content

$$AB \parallel EF, \qquad CD \parallel EF, \qquad \text{PointLine}(A, B) \neq \text{PointLine}(C, D)$$

imply

$$AB \parallel CD.$$

The proof does not pass through angle congruences.

Assume, for contradiction, that the first and third lines meet at a point P. Then both lines pass through P, and both are disjoint from EF.

Since P cannot lie on EF, there are two lines through the same point P, each parallel to EF.

Hilbert's Group IV says that through a point outside a line there is at most one line disjoint from the given line. Therefore the two lines through P must coincide.

Hence

$$\text{PointLine}(A, B) = \text{PointLine}(C, D),$$

contradicting the distinctness hypothesis.

Thus the first and third lines have no common point, and therefore

$$AB \parallel CD.$$

5 Lean Formalization

The reusable Hilbert-level theorem is

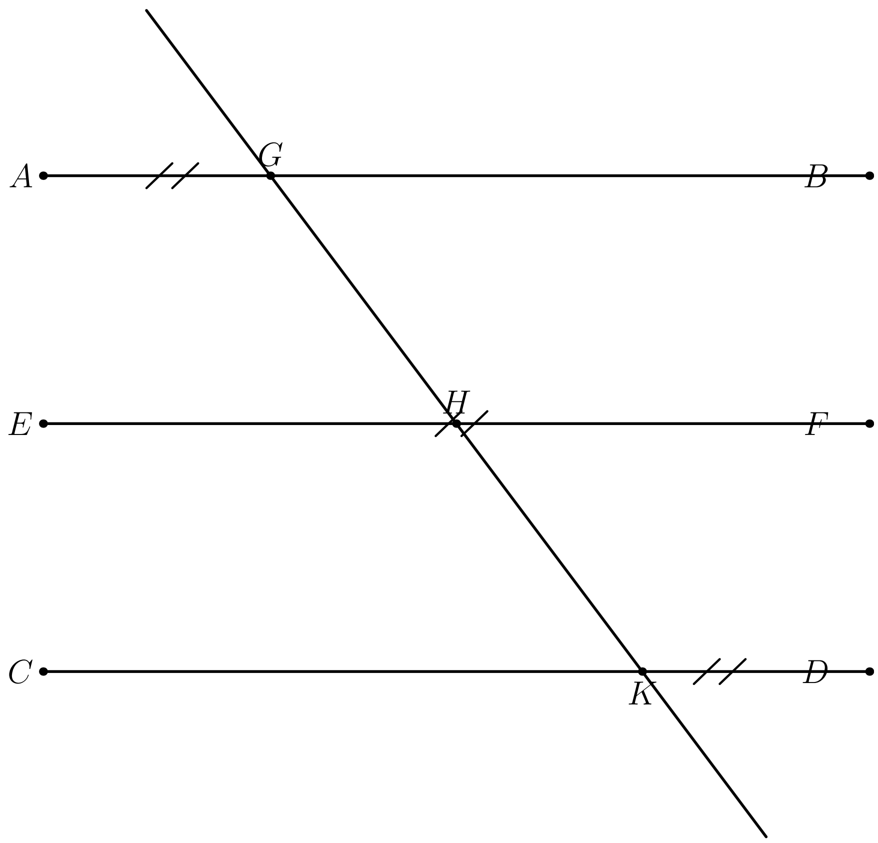

Figure 2: The Hilbert proof by contradiction. If the two outer lines met at P, they would be two distinct parallels through the same point to the middle line EF, contradicting Group IV.

```
theorem hilbert_parallel_transitive_distinct
    [HilbertEuclideanPlane Geo]
    (A B C D E F : Geo.Point)
    (hAB_EF : Geo.Parallel A B E F)
    (hCD_EF : Geo.Parallel C D E F)
    (hDistinct :
      Geo.PointLine A B != Geo.PointLine C D) :
    Geo.Parallel A B C D
```

The proof first obtains incidence lines through the three endpoint pairs:

$$AB, \qquad CD, \qquad EF.$$

The two hypotheses of parallelism are converted into disjointness of the corresponding incidence lines.

To prove $AB \parallel CD$, the proof assumes a common point P of the first and third point-line carriers. Membership in the carriers is transported to incidence on the chosen Hilbert lines.

The two parallel hypotheses imply that P is outside the line EF. The call

```
HilbertEuclideanPlane.parallel_unique
```

then identifies the two lines through P.

Finally,

```
hilbert_pointLine_eq_of_points_on_line
```

converts equality of the incidence carriers into equality of the project point-line carriers, contradicting `hDistinct`.

The Book I theorem

```
euclid_proposition_30
```

is consequently only a wrapper around this reusable Hilbert theorem.

6 Logical Position of Proposition I.30

The sequence I.27–I.30 now has a clear structure:

$$
\begin{aligned}
\text{I.27} &: \text{equal alternate angles} \longrightarrow \text{parallel},\\
\text{I.28} &: \text{other angle conditions} \longrightarrow \text{parallel},\\
\text{I.29} &: \text{parallel} \longrightarrow \text{angle relations},\\
\text{I.30} &: \text{parallel to the same line} \longrightarrow \text{parallel to one another}.
\end{aligned}
$$

The first two propositions are neutral.

Proposition I.29 introduces the Euclidean direction by using Hilbert's parallel axiom.

Proposition I.30 is then another direct manifestation of the same Euclidean principle: uniqueness of a parallel through a point.

Thus I.30 does not introduce a new independent parallel mechanism. It packages a structural consequence of Group IV.

7 Conclusion

Proposition I.30 becomes particularly transparent in the Hilbert reconstruction.

Euclid proves the theorem indirectly through angle relations:

$$\text{I.29} \longrightarrow \text{angle transitivity} \longrightarrow \text{I.27}.$$

The formal Hilbert development instead exposes the structural content:

$$\text{two distinct lines parallel to the same line} \longrightarrow \text{they cannot meet},$$

because any intersection point would produce two different parallels through one point to the same given line.

The resulting theorem

`hilbert_parallel_transitive_distinct`

is reusable beyond Proposition I.30, while

`euclid_proposition_30`

records the corresponding result in the sequence of Book I.